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<http://dx.doi.org/10.1016/j.jaad.2013.10.017>

Focal nail dystrophy in a cutaneous nerve distribution after trauma

To the Editor: Dystrophic nail changes have been described in association with traumatic injury, and may be focal or generalized. The determinants of the pattern of nail dystrophy remain uncertain, although immobilization is thought to be a major contributor. We report a case wherein unilateral nail dystrophy appeared to follow a median nerve distribution after a thumb injury suggesting a neurogenic etiology for its development.

A 10-year old schoolgirl sustained a closed oblique fracture to the proximal phalanx of her left thumb (Fig 1) following a crush injury in a hinged gate. She had no other injuries and at no stage demonstrated any clinical evidence of nerve injury or direct nail or fingertip damage. Initial swelling and edema were confined to the thumb and subsided with elevation and immobilization of the thumb, hand, and wrist in a plaster cast for 2 weeks. Her fingers were excluded from the cast and were freely mobile.

She developed dystrophic nail changes (Fig 2) affecting the thumb, index, and middle fingers of the left hand; these were first apparent approximately 2 to 3 weeks after the original injury and progressed distally from the base of the nails. There were no other associated symptoms or signs. She regained a full range of movement in all digits. The appearance of the nails had resolved completely by approximately 6 months after the injury.

Nail changes have been described in the literature in association with a variety of local and systemic conditions, as well as following injury. Localized nail dystrophy has been described in association with spinal trauma¹ and with reflex sympathetic dystrophy²; the presence of Beau lines is often observed. Open access under CC BY-NC-ND license.



Fig 1. Initial radiologic appearance showing fractured proximal phalanx of left thumb.



Fig 2. Dystrophic nail changes to thumb, index, and middle fingers of left hand. (Photograph obtained approximately 18 weeks after the initial injury.)

There is evidence that nail plate growth may be accelerated by trauma and is slowed by immobilization.³ While localized nail changes similar to those reported here have been described after direct nerve injury,⁴ we have been unable to find any reports of nail growth disruption of this pattern without evidence of nerve damage. Anatomic studies suggest that innervation of the nail beds is highly variable, but in most cases the nail beds probably receive dual sensory input via the proper digital and dorsal cutaneous nerves from both the radial and either median or ulnar nerve.⁵ The pattern seen in our case and that of Ross and Ward⁴ suggests that the volar innervation may dominate.

Our case highlights the association of trauma with nail dystrophy, and the distribution supports a neurogenic influence rather than immobilization in its development. We suggest that neurotrophic factors released locally by the relevant cutaneous sensory nerve may be important here.

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Funding sources: None.

Conflicts of interest: None declared.

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<http://dx.doi.org/10.1016/j.jaad.2013.10.021>

Dermatomyositis associated with capecitabine in the setting of malignancy

To the Editor: Dermatomyositis (DM) is a rare autoimmune condition characterized by cutaneous and muscle inflammation. In addition to malignancy, 24 medications have also been implicated in triggering DM.¹ Herein, we report a new case of DM triggered by oral capecitabine, a prodrug of 5-fluorouracil (5-FU), in the setting of underlying malignancy.

A 76-year-old man with a recent diagnosis of metastatic gastric adenocarcinoma presented with diffuse rash and muscle weakness 2 weeks after receiving a cycle of carboplatin with capecitabine (1 g orally twice daily). He had previously received 1 infusion of carboplatin alone 3 weeks before this cycle without any adverse effects. On examination, the patient was noted to have poikilodermatous plaques with shallow erosions scattered in a V-distribution on his upper chest and upper back (Fig 1). His hands contained erythematous, scaly papules symmetrically distributed over the metacarpophalangeal and interphalangeal joints (Gottron papules) as well as periungual erythema (Fig 1). Muscle strength testing revealed severe proximal weakness of the lower extremities, leading to an inability to ambulate. No electromyography (EMG) was conducted in this case.



Fig 1. Dermatomyositis, clinical images. Erosive erythematous patches on chest. Inset shows erythematous papules on interphalangeal joints (Gottron sign) and periungual erythema.

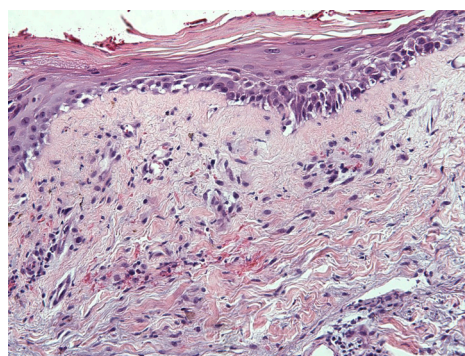


Fig 2. Dermatomyositis, histopathology. Punch biopsy of poikilodermatous erosive plaque on the shoulder shows an interface dermatitis with mild spongiosis, epidermal necrosis, and subepidermal vesicle formation. (Hematoxylin-eosin stain; original magnification $\times 20$.)

Laboratory analysis revealed an elevated creatine kinase (CK) of 2630 U/L, aldolase of 10.9 U/L, erythrocyte sedimentation rate (ESR) of 66 mm/hr, C-reactive protein (CRP) of 1.46 mg/dL, and lactate dehydrogenase (LDH) of 294 IU/L. Results were negative for antinuclear antibody (ANA), anti-Jo, and anti-centromere antibodies. A punch biopsy of the shoulder revealed an interface dermatitis with mild spongiosis, epidermal necrosis, and subepidermal vesicle formation (Fig 2). Thus, clinical, laboratory, and histopathologic findings were consistent with a diagnosis of DM. Capecitabine was discontinued and oral prednisone was initiated, resulting in reduction in CK level to 698 U/L and improved strength and ability to ambulate. The patient was discharged home. Due to the oncologic need to treat the patient's gastric adenocarcinoma, he was restarted